

**Module Name: ME4220 – Principal of Naval Architecture**

**Semester: 7**

## **CONCEPT DESIGN OF AN AMBULANCE CRAFT**

**By**

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## Table of Contents

|                                                    |    |
|----------------------------------------------------|----|
| Table of Contents.....                             | 1  |
| 1 Similar Craft Analysis.....                      | 5  |
| 1.1 Maximum Speed Vs Overall Length Selection..... | 5  |
| 1.2 Overall Length Vs Beam Selection.....          | 7  |
| 1.3 Beam Vs Draft Selection.....                   | 8  |
| 1.4 Draft Vs Displacement Selection.....           | 10 |
| 1.5 Displacement Vs Power Selection.....           | 11 |
| 1.6 Max Speed Vs Block Coefficient Selection.....  | 13 |
| 1.7 Final Values.....                              | 14 |
| 2 3D Hull Model.....                               | 15 |
| 2.1 Lines Plan.....                                | 15 |
| 2.2 3D Hull Development.....                       | 16 |
| 2.3 Solid Works Model.....                         | 18 |
| 3 Basic Hydrostatics and Intact Stability.....     | 19 |
| 4 General Arrangement Plan Drawings.....           | 21 |
| 4.1 General Specifications.....                    | 21 |
| 4.2 Accommodation and Layout.....                  | 22 |
| 4.2.1 Main Deck.....                               | 22 |
| 4.2.2 Lower Deck.....                              | 23 |
| 4.2.3 Functional and Medical Features.....         | 24 |
| 4.2.4 Amenities & Systems.....                     | 24 |

|       |                                             |                                     |
|-------|---------------------------------------------|-------------------------------------|
| 4.2.5 | Safety & Compliance .....                   | 25                                  |
| 5     | Weight and Weight Center Estimation .....   | 25                                  |
| 6     | Resistance and Power .....                  | 28                                  |
| 7     | Structural Arrangements .....               | <b>Error! Bookmark not defined.</b> |
| 8     | Subdivisions and Freeboard Comparison ..... | 32                                  |
| 9     | Stability assessment .....                  | <b>Error! Bookmark not defined.</b> |
| 10    | Cost Analysis .....                         | 33                                  |

## List of Figures

|                                                                  |    |
|------------------------------------------------------------------|----|
| Figure 1 : Summary of Similar Hull Analysis .....                | 5  |
| Figure 2 : Max Speed Vs Overall Length .....                     | 6  |
| Figure 3 : Max Speed Vs Overall Length Regression Line .....     | 6  |
| Figure 4 : Overall Length Vs Beam .....                          | 7  |
| Figure 5 : Overall Length Vs Beam Regression Line .....          | 8  |
| Figure 6 : Beam Vs Draft .....                                   | 9  |
| Figure 7 : Beam Vs Draft Regression Line .....                   | 9  |
| Figure 8 : Draft Vs Displacement .....                           | 10 |
| Figure 9 : Draft Vs Displacement Regression Line .....           | 11 |
| Figure 10 : Displacement Vs Power .....                          | 12 |
| Figure 11 : Displacement Vs Power Regression Line .....          | 12 |
| Figure 12 : Max Speed Vs Block Coefficient .....                 | 13 |
| Figure 13 : Max Speed Vs Block Coefficient Regression Line ..... | 14 |
| Figure 14 : Final Calculated Values .....                        | 15 |
| Figure 15 : Lines Plan of the hull .....                         | 15 |
| Figure 16 : Front view of the Hull .....                         | 16 |
| Figure 17 : Side View 1 of the Hull .....                        | 17 |
| Figure 18 : Side View 2 of the Hull .....                        | 17 |
| Figure 19 : Solid Works Model View 1 .....                       | 18 |
| Figure 20 : Solid Works Model View 2 .....                       | 18 |
| Figure 21 : Basic Hydrostatics Information .....                 | 19 |
| Figure 22 : Heel Angle vs righting arm and trim angle .....      | 20 |

|                                                                                                      |    |
|------------------------------------------------------------------------------------------------------|----|
| Figure 23 : Main Deck's GA Plan .....                                                                | 22 |
| Figure 24 : Lower Deck's GA Plan .....                                                               | 23 |
| Figure 25 : Weight centre Calculations .....                                                         | 26 |
| Figure 26 : Locations of centre of gravities .....                                                   | 26 |
| Figure 27: Centres of gravity .....                                                                  | 27 |
| Figure 28 : Speed vs Resistance power and Power table .....                                          | 28 |
| Figure 29 : Power vs Speed graph .....                                                               | 29 |
| Figure 30 : Wave pattern visualization (using MAXSURF Resistance CONNECT Edition x64 software) ..... | 30 |
| Figure 31 : 600 hp inboard Cummins marine engine .....                                               | 31 |
| Figure 32 : Key attributes of the engine.....                                                        | 31 |
| Figure 33 : Front View of Bulkheads .....                                                            | 32 |
| Figure 34 : Bulkheads Designs .....                                                                  | 33 |

## List of Tables

|                                |    |
|--------------------------------|----|
| Table 1 : Cost Estimation..... | 34 |
|--------------------------------|----|

# 1 Similar Craft Analysis

An analysis of existing ambulance and rescue craft was conducted using data from comparable vessels currently in operation. The reviewed craft typically fall within an overall length range of approximately 9–23 m, with displacements varying from about 5.5 to 45 tonnes, depending on hull type and operational role. Most vessels operate at maximum speeds between 18 and 33 knots, indicating the importance of high-speed capability for emergency response missions. Monohull configurations are commonly used for smaller and medium-sized craft due to their simplicity and efficiency, while larger ambulance vessels often adopt catamaran hulls to improve deck area, transverse stability, and patient comfort. The block coefficients generally lie between 0.32 and 0.45, reflecting slender hull forms optimized for high-speed operation. Propulsive power requirements range from approximately 110 kW to 750 kW, increasing with displacement and target speed. These observed trends were used to validate the selected principal dimensions, hull form, and powering estimates of the proposed craft, ensuring consistency with proven designs and realistic performance expectations.

| No. | Anadolu Shipyard Ambulance Boat   | L overall (m) | B overall (m) | T (max. displ. at rest under hull)(m) | Δ (Displacement FULL) (tonnes) | Max Speed (V) (knots) | Propulsive Power (P) (HP/kW) | CB (Block Coeff.) | Hull Type             | References                                                                                                        |
|-----|-----------------------------------|---------------|---------------|---------------------------------------|--------------------------------|-----------------------|------------------------------|-------------------|-----------------------|-------------------------------------------------------------------------------------------------------------------|
| 1   | Icarus Marine 15.6m Ambulance     | 15.6          | 5.9           | 0.7                                   | 19.2                           | 33                    | 2 x 375 kW                   | N/A               | Monohull              | <a href="https://www.icasmarine.com/p/156m-ambulance-boat/">https://www.icasmarine.com/p/156m-ambulance-boat/</a> |
| 2   | Kewatec Ambulance 1370            | 13.7          | 3.8           | 1.2                                   | N/A                            | 35                    | 2 x 400 – 600 HP             | N/A               | Monohull (Aluminum)   |                                                                                                                   |
| 3   | Anadolu Shipyard Ambulance Boat   | 16.45         | 6             | 1.2                                   | N/A                            | 30                    | 2 x 900 BHP                  | N/A               | Catamaran             | <a href="https://www.anadolushipyard.com/boats/">https://www.anadolushipyard.com/boats/</a>                       |
| 4   | Kewatec Ambulance 2280 CAT        | 22.8          | 7.6           | 1.2                                   | N/A                            | 25 – 35               | 2 x 700 – 1000 HP            | N/A               | Catamaran             |                                                                                                                   |
| 5   | Water Ambulance (East Kalimantan) | 8.81          | 2.65          | 0.45                                  | N/A                            | 18                    | 150 HP                       | 0.32              | Monohull              |                                                                                                                   |
| 6   | Neil Marine 10.5m Ambulance       | 10.77         | 3.1           | 0.75                                  | 4.6 (Light)                    | N/A                   | N/A                          | N/A               | Monohull (FRP)        |                                                                                                                   |
| 7   | New Build 11m Ambulance/SAR       | 11            | 3.8           | 0.62                                  | 6.2 (Approx.)                  | N/A                   | N/A                          | N/A               | Monohull (K-TOUGH)    |                                                                                                                   |
| 8   | Hukkelberg SAR Boat 1400          | 14            | 4.3           | 1.2                                   | N/A                            | 36 (Approx.)          | N/A                          | N/A               | Monohull (Diesel IPS) |                                                                                                                   |
| 9   | SG900 Aluminium Speed Boat (9m)   | 9             | 2.2 – 2.5     | N/A                                   | N/A                            | 25 – 39               | 90 – 200 HP                  | N/A               | Monohull (Aluminum)   |                                                                                                                   |
| 10  | Icarus Marine 13m SAR Catamaran   | 13            | 4.8           | 0.8                                   | 12                             | 45                    | 4 x 350 HP (OBM)             | N/A               | Catamaran             |                                                                                                                   |
| 11  | Icarus Marine 13m SAR Hydrofoil   | 13            | 4.2           | 0.76                                  | 10.4                           | 50                    | N/A                          | N/A               | Catamaran/Hydrofoil   |                                                                                                                   |
| 12  | Icarus Marine 15m Fire Fighting   | 15.1          | 5             | 0.85                                  | 18.2                           | 40                    | 2 x 599 kW                   | N/A               | Monohull              |                                                                                                                   |

**Figure 1 : Summary of Similar Hull Analysis**

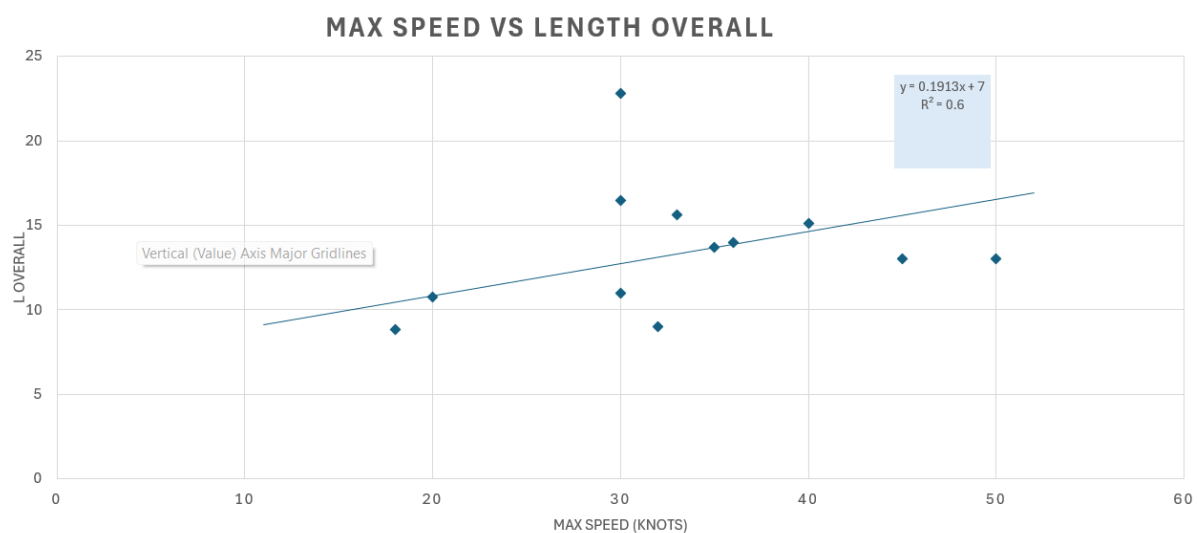
## 1.1 Maximum Speed Vs Overall Length Selection

An analysis of similar high-speed craft was carried out by examining the relationship between maximum speed and length overall using data collected from existing vessels. A linear regression was performed, which showed a moderate correlation between speed and length, with an  $R^2$  value of 0.60. The derived relationship, with a slope of approximately 0.19 and an intercept of 7.0, indicates that higher operating speeds generally require an increase in vessel length to ensure hydrodynamic efficiency, stability, and acceptable powering demands. The predicted lengths fall within reasonable upper and lower confidence limits, confirming that the dataset represents realistic and proven craft designs.

Based on this analysis, a design speed of **30 knots** was selected as it aligns well with the operational requirements of a high-speed ambulance craft while remaining consistent with similar vessels. Using the regression relationship, the corresponding length overall was estimated to be approximately **12.7 m**, which lies within the acceptable prediction band. This selected speed length combination was therefore adopted for the proposed design, ensuring that the vessel performance is benchmarked against existing craft and supported by statistical analysis rather than arbitrary assumptions.

| Max speed (V) | Length overall ( L) | L predicted | Residuals | Upper limit | Lower limit |
|---------------|---------------------|-------------|-----------|-------------|-------------|
| 33            | 15.6                | 13.3129     | 2.2871    | 14.5129     | 12.1129     |
| 35            | 13.7                | 13.6955     | 0.0045    | 14.8955     | 12.4955     |
| 30            | 16.45               | 12.739      | 3.711     | 13.939      | 11.539      |
| 30            | 22.8                | 12.739      | 10.061    | 13.939      | 11.539      |
| 18            | 8.81                | 10.4434     | -1.6334   | 11.6434     | 9.2434      |
| 20            | 10.77               | 10.826      | -0.056    | 12.026      | 9.626       |
| 30            | 11                  | 12.739      | -1.739    | 13.939      | 11.539      |
| 36            | 14                  | 13.8868     | 0.1132    | 15.0868     | 12.6868     |
| 32            | 9                   | 13.1216     | -4.1216   | 14.3216     | 11.9216     |
| 45            | 13                  | 15.6085     | -2.6085   | 16.8085     | 14.4085     |
| 50            | 13                  | 16.565      | -3.565    | 17.765      | 15.365      |
| 40            | 15.1                | 14.652      | 0.448     | 15.852      | 13.452      |

**Figure 2 : Max Speed Vs Overall Length**



**Figure 3 : Max Speed Vs Overall Length Regression Line**

|                    |      |
|--------------------|------|
| Slope              | 0.19 |
| Intercept          | 7.00 |
| Standard Deviation | 3.84 |
| $R^2$              | 0.60 |

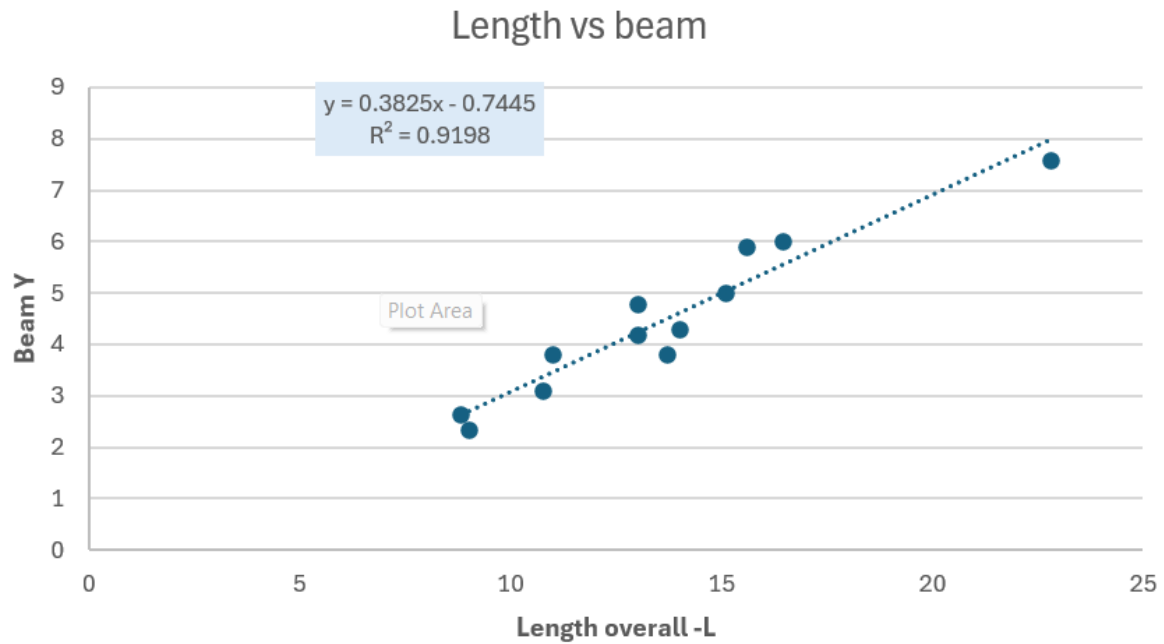
## 1.2 Overall Length Vs Beam Selection

A similar regression-based analysis was carried out to examine the relationship between length overall and beam using data from comparable high-speed craft. The results show a strong linear correlation between these two parameters, with an  $R^2$  value of 0.92, indicating that beam scales consistently with vessel length for this class of craft. The derived relationship, with a slope of approximately 0.38 and a small negative intercept, reflects typical proportions adopted in practice to balance transverse stability, deck space, and hydrodynamic efficiency.

Using the previously selected length overall of approximately 12.7 m, the corresponding beam was estimated from the regression model to be about **4.1 m**. This value lies well within the upper and lower prediction limits obtained from the dataset, confirming its suitability. The selected beam therefore represents a realistic and proven proportion, ensuring adequate stability and internal space while maintaining acceptable resistance characteristics for high-speed ambulance craft operations.

| Length overall L | Beam Y | Beam Y predicted | Residuals | Upper limit | Lower limit |
|------------------|--------|------------------|-----------|-------------|-------------|
| 15.6             | 5.9    | 5.2225           | 0.6775    | 7.0621      | 3.3829      |
| 13.7             | 3.8    | 4.49575          | -0.69575  | 6.33535     | 2.65615     |
| 16.45            | 6      | 5.547625         | 0.452375  | 7.387225    | 3.708025    |
| 22.8             | 7.6    | 7.9765           | -0.3765   | 9.8161      | 6.1369      |
| 8.81             | 2.65   | 2.625325         | 0.024675  | 4.464925    | 0.785725    |
| 10.77            | 3.1    | 3.375025         | -0.275025 | 5.214625    | 1.535425    |
| 11               | 3.8    | 3.463            | 0.337     | 5.3026      | 1.6234      |
| 14               | 4.3    | 4.6105           | -0.3105   | 6.4501      | 2.7709      |
| 9                | 2.35   | 2.698            | -0.348    | 4.5376      | 0.8584      |
| 13               | 4.8    | 4.228            | 0.572     | 6.0676      | 2.3884      |
| 13               | 4.2    | 4.228            | -0.028    | 6.0676      | 2.3884      |
| 15.1             | 5      | 5.03125          | -0.03125  | 6.87085     | 3.19165     |
| 12.739           |        | 4.1281675        |           | 5.9677675   | 2.2885675   |

**Figure 4 : Overall Length Vs Beam**



**Figure 5 : Overall Length Vs Beam Regression Line**

|                    |         |
|--------------------|---------|
| Slope              | 0.38    |
| Intercept          | -0.74   |
| Standard Deviation | #DIV/0! |
| $R^2$              | 0.92    |

### 1.3 Beam Vs Draft Selection

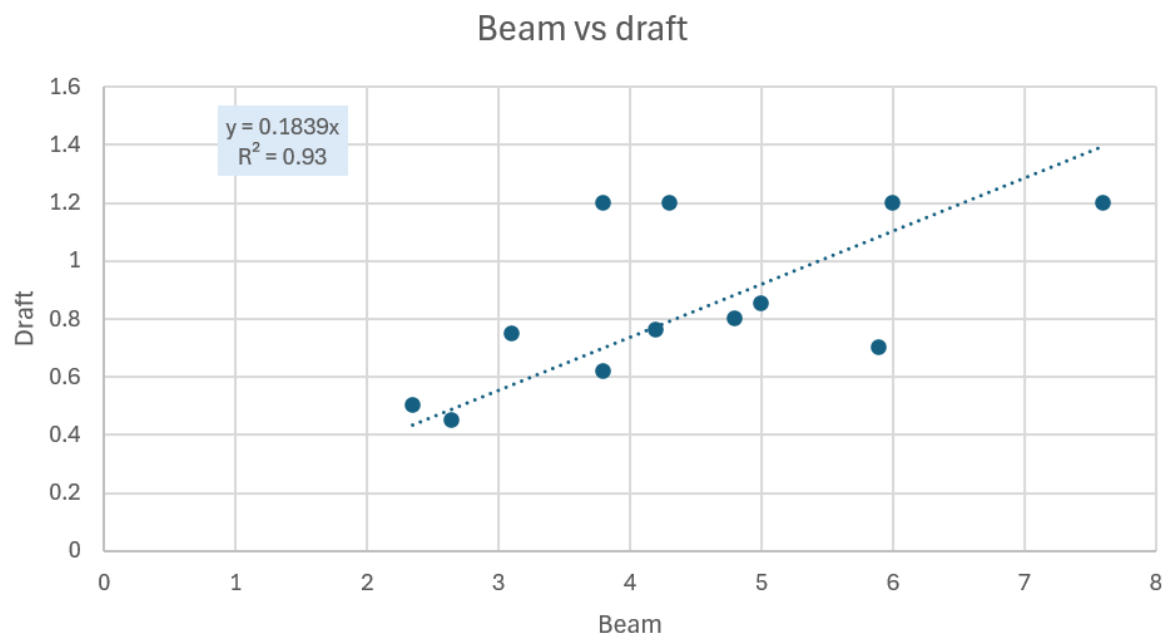
The relationship between beam and draft was examined using data from similar high-speed craft to determine a realistic draft for the proposed design. The regression analysis shows a strong linear correlation, with an  $R^2$  value of 0.93, indicating that draft increases proportionally with beam for vessels in this category. The obtained slope of approximately 0.18 reflects typical design practice, where wider craft require slightly deeper drafts to maintain adequate displacement, stability, and structural efficiency.

Using the previously selected beam of approximately 4.13 m, the corresponding draft was estimated from the regression model to be about **0.76 m**. This value lies well within the upper and lower prediction limits derived from the dataset, confirming that the selected draft is consistent with existing high-speed ambulance and rescue craft. The adopted draft therefore

ensures sufficient underwater volume and stability while retaining shallow-water operability, which is essential for nearshore and emergency response missions.

| Beam  | Draft | Expected draft | Residuals | Upper limit | Lower limit |
|-------|-------|----------------|-----------|-------------|-------------|
| 5.9   | 0.7   | 1.08501        | -0.38501  | 2.94501     | -0.77499    |
| 3.8   | 1.2   | 0.69882        | 0.50118   | 2.55882     | -1.16118    |
| 6     | 1.2   | 1.1034         | 0.0966    | 2.9634      | -0.7566     |
| 7.6   | 1.2   | 1.39764        | -0.19764  | 3.25764     | -0.46236    |
| 2.65  | 0.45  | 0.487335       | -0.037335 | 2.347335    | -1.372665   |
| 3.1   | 0.75  | 0.57009        | 0.17991   | 2.43009     | -1.28991    |
| 3.8   | 0.62  | 0.69882        | -0.07882  | 2.55882     | -1.16118    |
| 4.3   | 1.2   | 0.79077        | 0.40923   | 2.65077     | -1.06923    |
| 2.35  | 0.5   | 0.432165       | 0.067835  | 2.292165    | -1.427835   |
| 4.8   | 0.8   | 0.88272        | -0.08272  | 2.74272     | -0.97728    |
| 4.2   | 0.76  | 0.77238        | -0.01238  | 2.63238     | -1.08762    |
| 5     | 0.85  | 0.9195         | -0.0695   | 2.7795      | -0.9405     |
| 4.128 |       | 0.7591392      |           | 2.6191392   | -1.1008608  |

**Figure 6 : Beam Vs Draft**



**Figure 7 : Beam Vs Draft Regression Line**

|                    |      |
|--------------------|------|
| Slope              | 0.18 |
| Intercept          | 0.00 |
| Standard Deviation | 0.24 |
| $R^2$              | 0.93 |

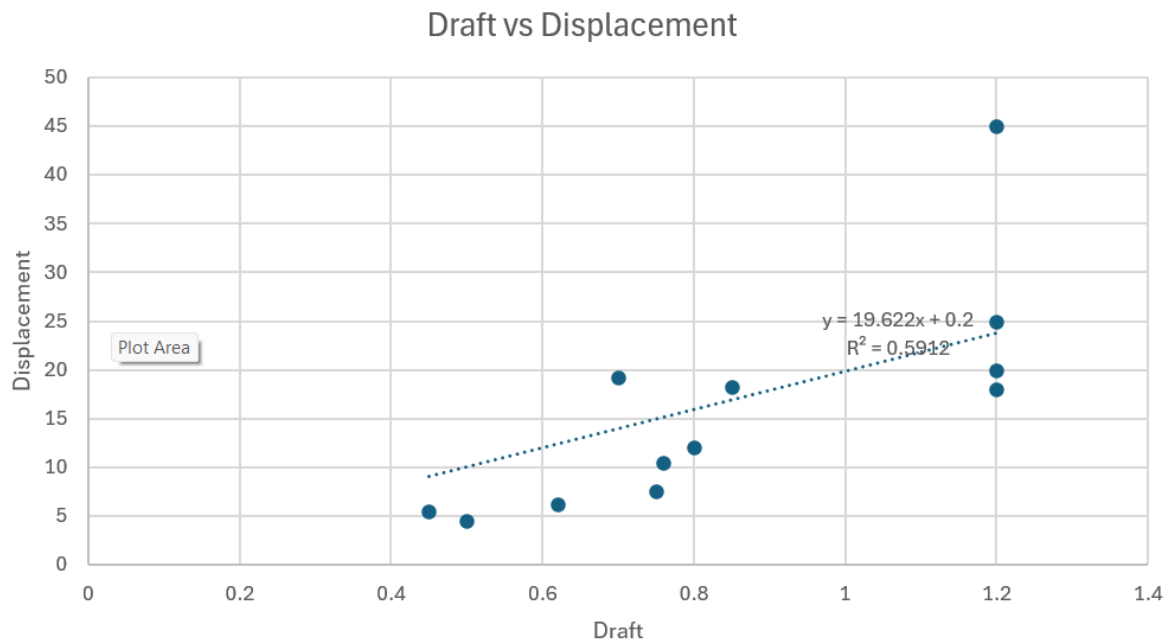
## 1.4 Draft Vs Displacement Selection

The relationship between draft and displacement was analyzed using data from similar high-speed craft to estimate a realistic displacement for the proposed vessel. The regression analysis indicates a moderate correlation between these parameters, with an  $R^2$  value of 0.59, which is reasonable given the influence of additional factors such as hull form, beam, and block coefficient on displacement. The obtained slope of approximately 19.6 shows that small increases in draft result in significant changes in displacement for vessels in this size and speed range.

Using the previously selected draft of approximately 0.76 m, the corresponding displacement was estimated from the regression model to be about 15.1 tones. This predicted value lies within the upper and lower confidence limits derived from the dataset, confirming that the displacement is consistent with comparable high-speed ambulance and rescue craft. The selected displacement was therefore adopted for subsequent hydrostatic, stability, and powering calculations, ensuring that the vessel's weight and buoyancy characteristics are aligned with proven reference designs.

| Draft | Displacement | Expected displacement | Residuals | Upper limit | Lower limit |
|-------|--------------|-----------------------|-----------|-------------|-------------|
| 0.7   | 19.2         | 13.9354               | 5.2646    | 15.1178     | 12.753      |
| 1.2   | 18           | 23.7464               | -5.7464   | 24.9288     | 22.564      |
| 1.2   | 25           | 23.7464               | 1.2536    | 24.9288     | 22.564      |
| 1.2   | 45           | 23.7464               | 21.2536   | 24.9288     | 22.564      |
| 0.45  | 5.5          | 9.0299                | -3.5299   | 10.2123     | 7.8475      |
| 0.75  | 7.5          | 14.9165               | -7.4165   | 16.0989     | 13.7341     |
| 0.62  | 6.2          | 12.36564              | -6.16564  | 13.54804    | 11.18324    |
| 1.2   | 20           | 23.7464               | -3.7464   | 24.9288     | 22.564      |
| 0.5   | 4.5          | 10.011                | -5.511    | 11.1934     | 8.8286      |
| 0.8   | 12           | 15.8976               | -3.8976   | 17.08       | 14.7152     |
| 0.76  | 10.4         | 15.11272              | -4.71272  | 16.29512    | 13.93032    |
| 0.85  | 18.2         | 16.8787               | 1.3213    | 18.0611     | 15.6963     |
| 0.759 |              | 15.093098             |           | 16.275498   | 13.910698   |

**Figure 8 : Draft Vs Displacement**



**Figure 9 : Draft Vs Displacement Regression Line**

|                    |       |
|--------------------|-------|
| Slope              | 19.62 |
| Intercept          | 0.20  |
| Standard Deviation | 5.78  |
| $R^2$              | 0.59  |

### 1.5 Displacement Vs Power Selection

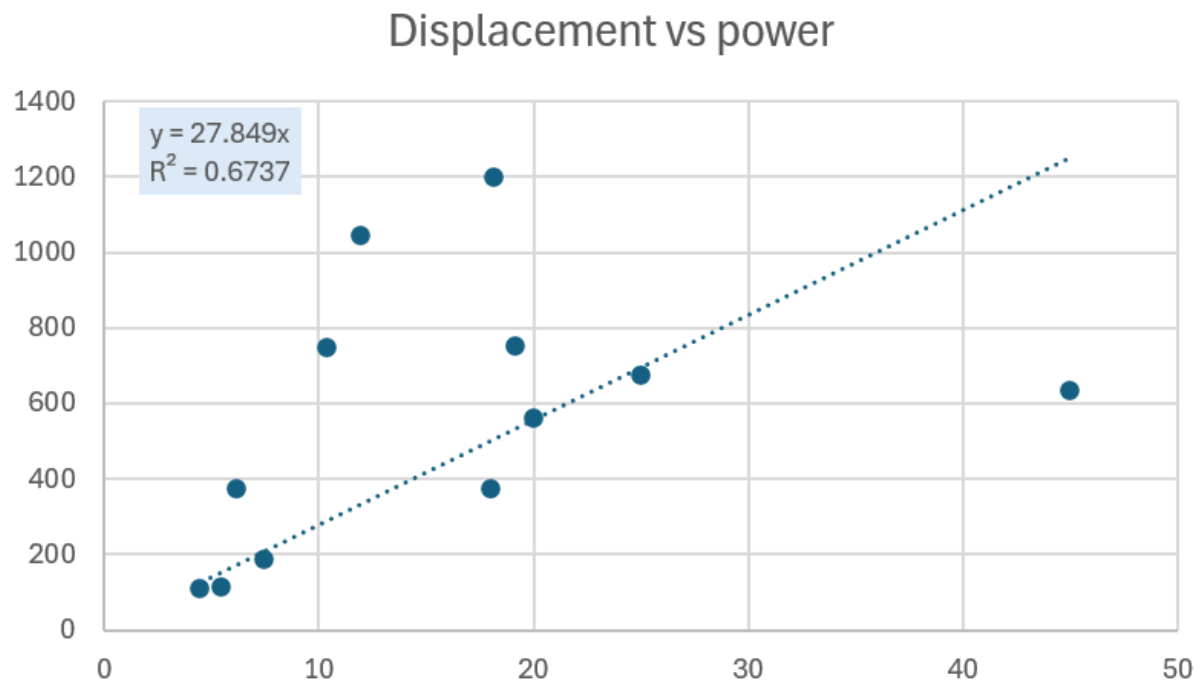
The relationship between vessel displacement and installed power was analyzed using data from similar high-speed craft to obtain a realistic estimate of the propulsion power requirement. The regression analysis shows a reasonable correlation between displacement and power, with an  $R^2$  value of 0.67, indicating that displacement is a key driver of power demand, although other factors such as hull form, speed, and propulsion type also influence the final requirement. The obtained slope of approximately 27.9 reflects the rapid increase in required power as displacement increases for high-speed operation.

Using the previously estimated displacement of approximately 15.1 tones, the corresponding power requirement was predicted to be about 420 kW based on the regression model. This value lies within the calculated prediction limits and is consistent with the power levels

observed in comparable high-speed ambulance and rescue craft. The estimated power was therefore adopted as a baseline for propulsion system selection and further refined during the resistance and powering analysis to account for service conditions, safety margins, and operational requirements.

| Displacement | Power | Estimated Power | Residuals | Upper limit | Lower limit |
|--------------|-------|-----------------|-----------|-------------|-------------|
| 19.2         | 750   | 534.7008        | 215.2992  | 536.0482    | 533.3534    |
| 18           | 373   | 501.282         | -128.282  | 502.6294    | 499.9346    |
| 25           | 671   | 696.225         | -25.225   | 697.5724    | 694.8776    |
| 45           | 634   | 1253.205        | -619.205  | 1254.5524   | 1251.8576   |
| 5.5          | 112   | 153.1695        | -41.1695  | 154.5169    | 151.8221    |
| 7.5          | 186   | 208.8675        | -22.8675  | 210.2149    | 207.5201    |
| 6.2          | 373   | 172.6638        | 200.3362  | 174.0112    | 171.3164    |
| 20           | 560   | 556.98          | 3.02      | 558.3274    | 555.6326    |
| 4.5          | 108   | 125.3205        | -17.3205  | 126.6679    | 123.9731    |
| 12           | 1044  | 334.188         | 709.812   | 335.5354    | 332.8406    |
| 10.4         | 746   | 289.6296        | 456.3704  | 290.977     | 288.2822    |
| 18.2         | 1198  | 506.8518        | 691.1482  | 508.1992    | 505.5044    |
| 15.093       |       | 420.324957      |           | 421.672357  | 418.977557  |

**Figure 10 : Displacement Vs Power**



**Figure 11 : Displacement Vs Power Regression Line**

|                    |        |
|--------------------|--------|
| Slope              | 27.85  |
| Intercept          | 0.00   |
| Standard Deviation | 330.66 |
| $R^2$              | 0.67   |

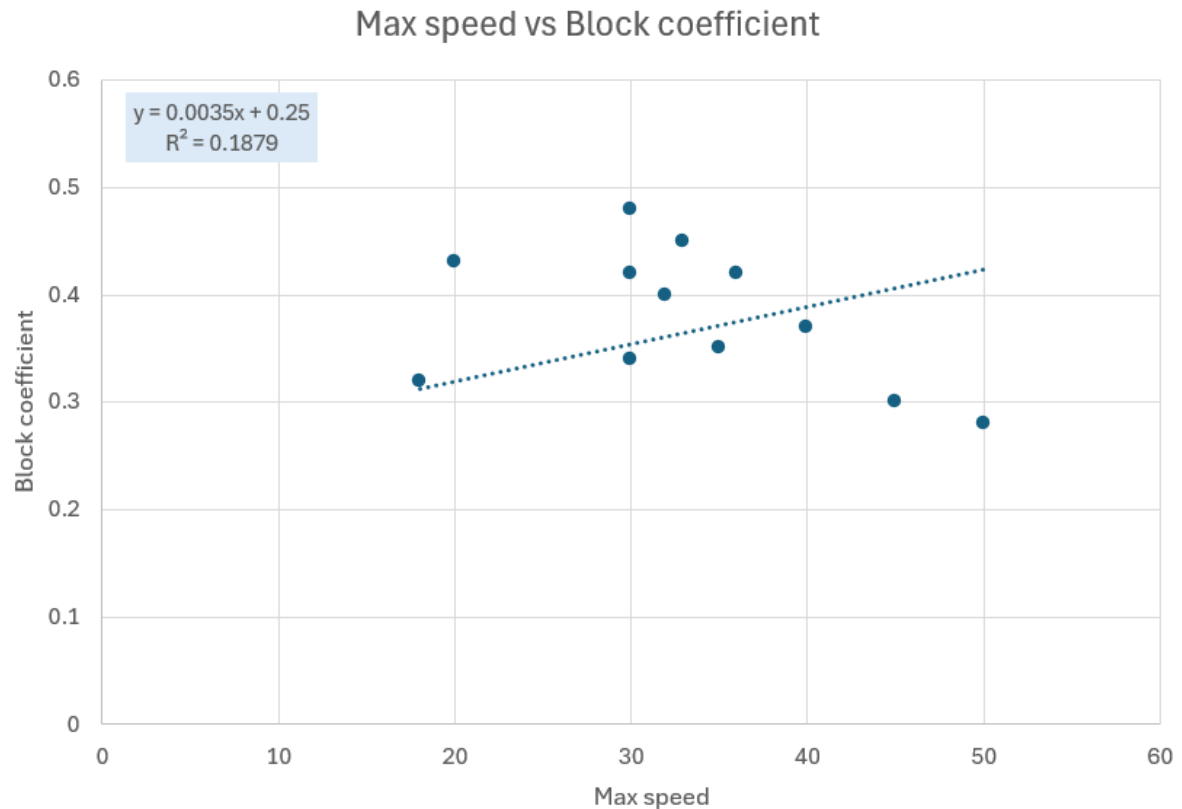
## 1.6 Max Speed Vs Block Coefficient Selection

The relationship between maximum speed and block coefficient was analyzed using data from similar high-speed craft to identify a suitable hull fullness for the proposed design. The regression analysis indicates a weak correlation between these two parameters, with an  $R^2$  value of 0.19, suggesting that block coefficient is not governed by speed alone but is also influenced by hull form, operational role, and design philosophy. Nevertheless, the trend shows that high-speed craft generally adopt relatively low to moderate block coefficients to reduce resistance and support efficient planning or semi-planning behavior.

For the selected design speed of 30 knots, the regression model predicts a block coefficient of approximately 0.35. This value lies within the observed range of comparable vessels and within the prediction limits obtained from the dataset. Considering the need for a balance between hydrodynamic efficiency and sufficient internal volume for medical spaces and equipment, a block coefficient in this range was adopted for the proposed craft. This selection ensures compatibility with high-speed performance requirements while maintaining practical space and weight distribution.

| Max speed | Block coefficient | Estimated block coefficient | Residuals | Upper limit | Lower limit |
|-----------|-------------------|-----------------------------|-----------|-------------|-------------|
| 33        | 0.45              | 0.3655                      | 0.0845    | 0.7413      | -0.0103     |
| 35        | 0.35              | 0.3725                      | -0.0225   | 0.7483      | -0.0033     |
| 30        | 0.34              | 0.355                       | -0.015    | 0.7308      | -0.0208     |
| 30        | 0.42              | 0.355                       | 0.065     | 0.7308      | -0.0208     |
| 18        | 0.32              | 0.313                       | 0.007     | 0.6888      | -0.0628     |
| 20        | 0.43              | 0.32                        | 0.11      | 0.6958      | -0.0558     |
| 30        | 0.48              | 0.355                       | 0.125     | 0.7308      | -0.0208     |
| 36        | 0.42              | 0.376                       | 0.044     | 0.7518      | 0.0002      |
| 32        | 0.4               | 0.362                       | 0.038     | 0.7378      | -0.0138     |
| 45        | 0.3               | 0.4075                      | -0.1075   | 0.7833      | 0.0317      |
| 50        | 0.28              | 0.425                       | -0.145    | 0.8008      | 0.0492      |
| 40        | 0.37              | 0.39                        | -0.02     | 0.7658      | 0.0142      |

**Figure 12 : Max Speed Vs Block Coefficient**



**Figure 13 : Max Speed Vs Block Coefficient Regression Line**

|                    |      |
|--------------------|------|
| Slope              | 0.00 |
| Intercept          | 0.25 |
| Standard Deviation | 0.03 |
| $R^2$              | 0.19 |

## 1.7 Final Values

The final design parameters were established by refining the values obtained from the similar craft analysis and applying practical design limits. The selected parameters ensure that the vessel meets its operational requirements while maintaining adequate stability, efficiency, and safety margins. Adjustments were made to account for layout needs, service conditions, and design robustness, resulting in a balanced configuration that is consistent with proven high-speed ambulance craft designs and suitable for subsequent detailed design stages.

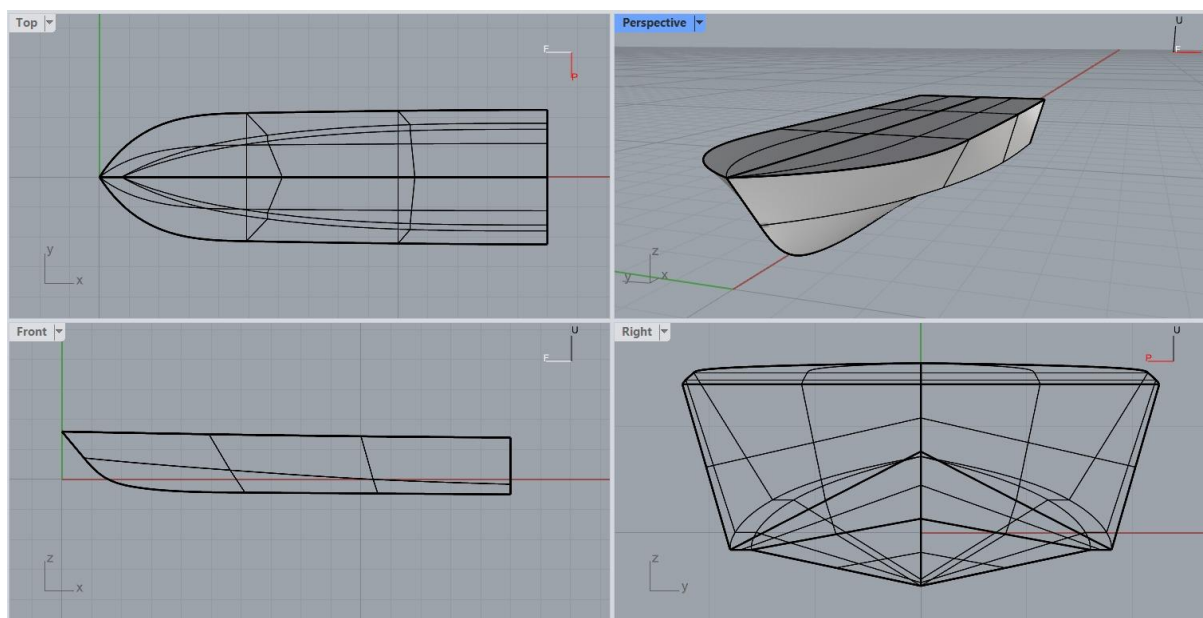
| Design Specifications        | Obtained Value from Similar Ship Analysis | Final Values After Limits |
|------------------------------|-------------------------------------------|---------------------------|
| Maximum speed (Knots)        | 30                                        | 30                        |
| Overall length (m)           | 12.739                                    | 15                        |
| Beam (m)                     | 4.128                                     | 5                         |
| Draft (m)                    | 0.759                                     | 1                         |
| Displacement (Tonnes)        | 15.093                                    | 15                        |
| Power (kW)                   | 421.67                                    | 450                       |
| Block coefficient            | 0.355                                     | 0.355                     |
| Calculated block coefficient | 0.368922297                               | 0.355                     |

**Figure 14 : Final Calculated Values**

## 2 3D Hull Model

The three-dimensional hull model was developed using the selected principal dimensions to accurately represent the vessel geometry. The model was created in Rhino to ensure smooth hull fairing and correct definition of waterlines, sections, and buttocks. This 3D hull form was then used as the basis for hydrostatic and stability evaluations in Orca3D, supporting reliable assessment of the vessel's performance and behaviour in service conditions.

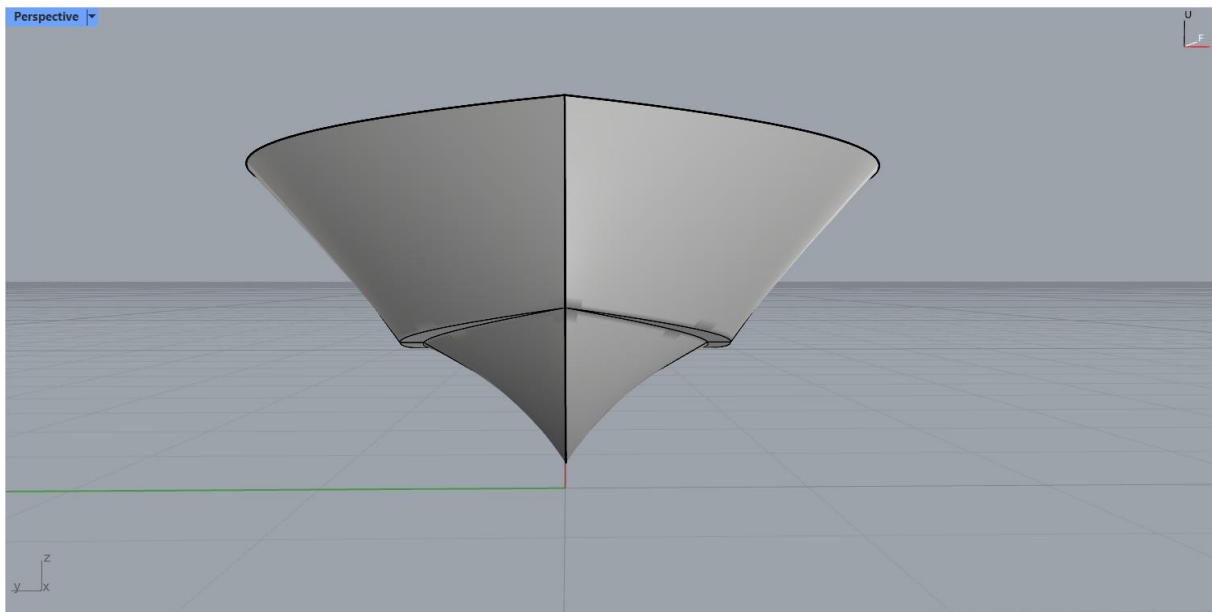
### 2.1 Lines Plan



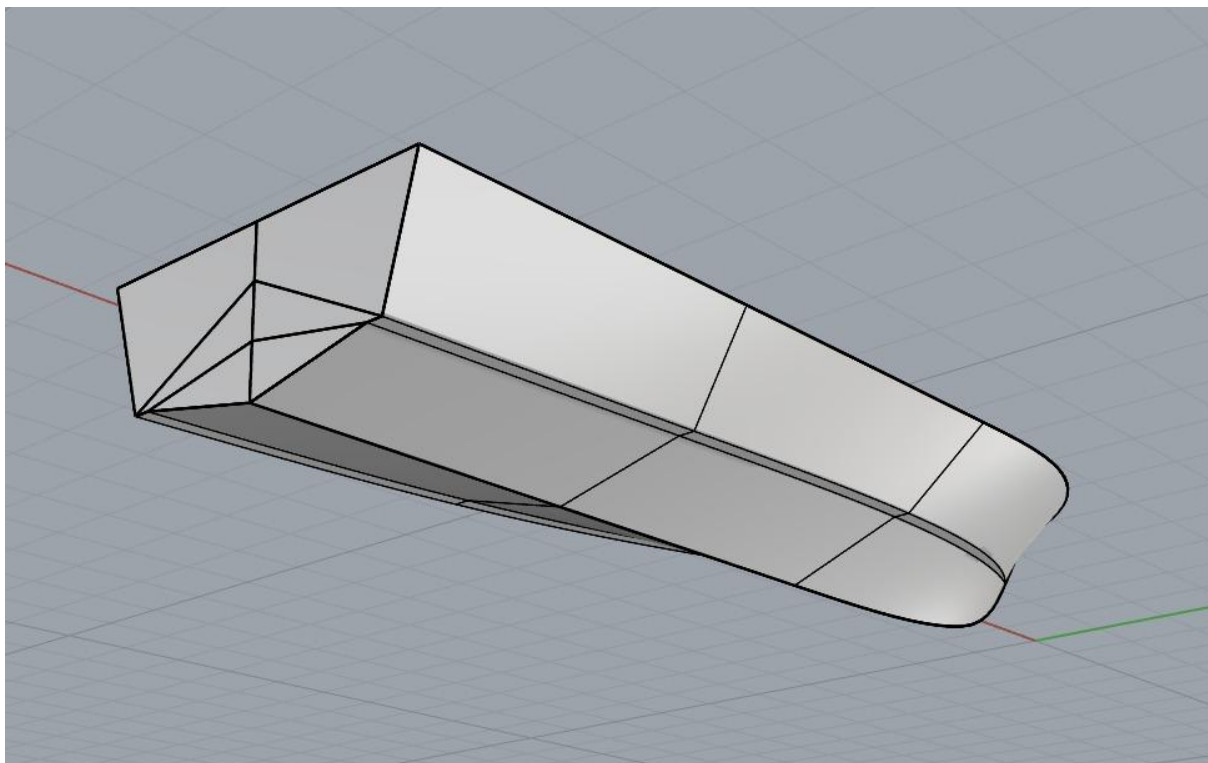
**Figure 15 : Lines Plan of the hull**

## 2.2 3D Hull Development

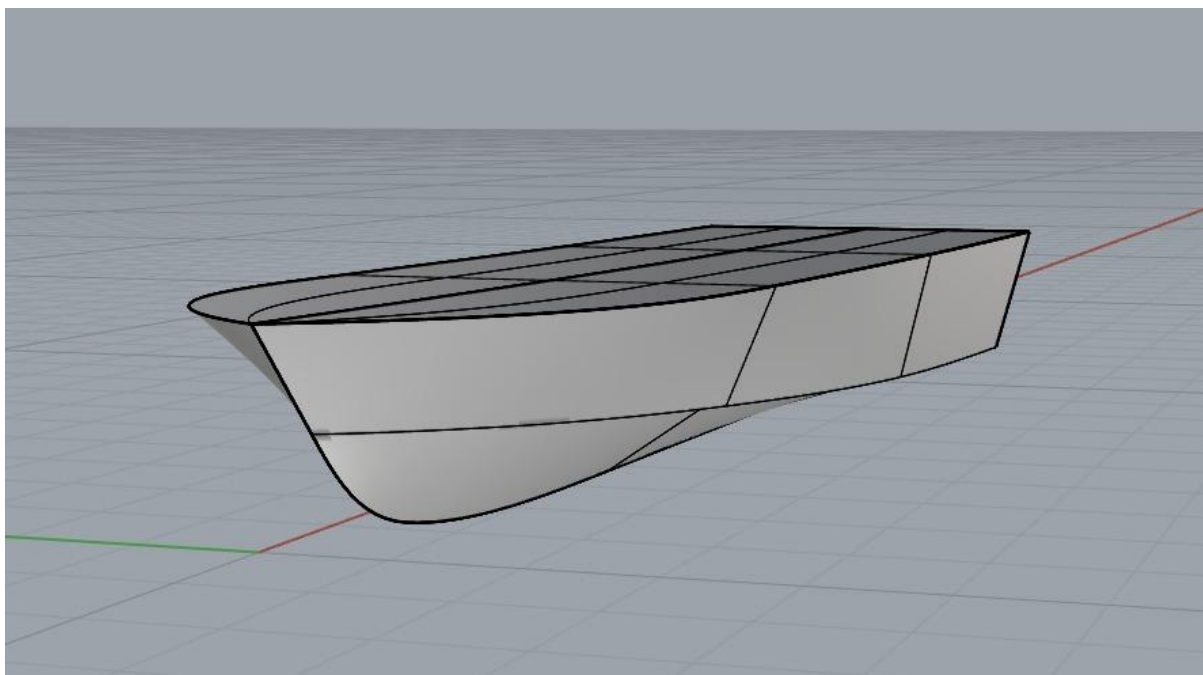
The hull geometry was further refined through detailed 3D development to achieve smooth surface continuity and realistic structural form. Key features such as the bow shape, deadrise variation, and transom geometry were adjusted to support high-speed performance and good seakeeping. This finalized 3D hull model serves as the foundation for stability analysis, resistance estimation, and subsequent arrangement and structural design.



**Figure 16 : Front view of the Hull**



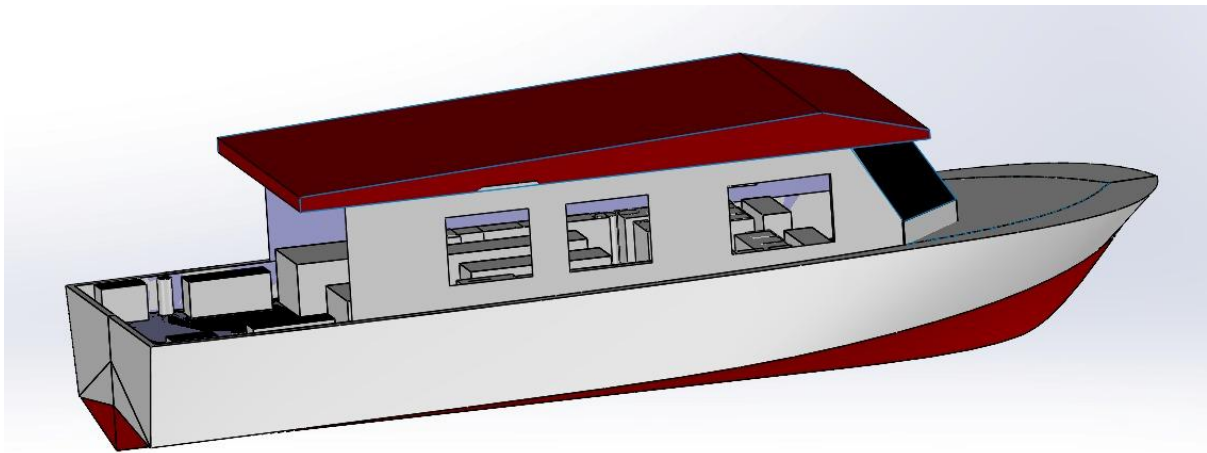
**Figure 17 : Side View 1 of the Hull**



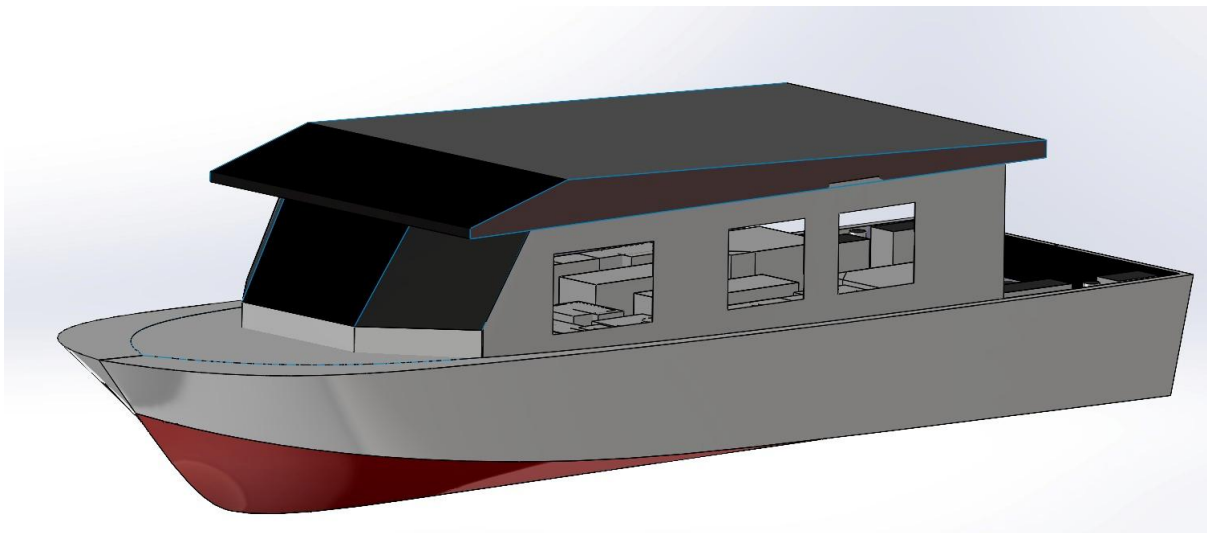
**Figure 18 : Side View 2 of the Hull**

### 2.3 Solid Works Model

The superstructure was designed using SolidWorks to accurately model the external geometry and internal arrangement. This allowed precise control of dimensions, clear definition of openings and glazing, and proper integration with the hull. The SolidWorks model supports layout validation, visualization, and further structural and weight assessments of the superstructure.



**Figure 19 : Solid Works Model View 1**



**Figure 20 : Solid Works Model View 2**

### 3 Basic Hydrostatics and Intact Stability

Using the Orca 3D add-in for Rhino 3D, the essential hydrostatic data for the hull were generated.

#### Default Project

Equilibrium Stability Analysis with Righting Arm (with fixed load or flotation plane)

Default Company

12/8/2025 1:40:11 PM

Orca3D Version 3.1.7 (4/8/2025)

C:\Users\acer\Downloads\hull001.3dm

Orca3D Design: No Orca3D Design

#### Hydrostatics Summary

| Load Case Input Parameters |                |           |           |          |
|----------------------------|----------------|-----------|-----------|----------|
| Load Case                  | Weight/Sinkage | LCG/Trim  | TCG/Heel  | VCG (mm) |
| Free-float Condition       | 15000.000 kgf  | 0.500 deg | 0.000 deg | 1000.000 |

| Resulting Model Attitude and Hydrostatic Properties |              |            |            |          |
|-----------------------------------------------------|--------------|------------|------------|----------|
| Load Case                                           | Sinkage (mm) | Trim (deg) | Heel (deg) | Ax (m^2) |
| Free-float Condition                                | 210.735      | 0.500BUp   | 0.000      | 0.000    |

| Load Case            | Displacement (kgf) | LCB (mm)  | TCB (mm) | VCB (mm) | Wet Area (m^2) |
|----------------------|--------------------|-----------|----------|----------|----------------|
| Free-float Condition | 14999.995          | 10008.219 | 0.000    | 94.790   | 50.097         |

| Load Case            | Awp (m^2) | LCF (mm) | TCF (mm) | VCF (mm) |
|----------------------|-----------|----------|----------|----------|
| Free-float Condition | 44.444    | 9455.272 | 0.000    | 293.258  |

| Load Case            | BM <sub>t</sub> (mm) | BM <sub>I</sub> (mm) | GM <sub>T</sub> (mm) | GM <sub>L</sub> (mm) |
|----------------------|----------------------|----------------------|----------------------|----------------------|
| Free-float Condition | 3656.960             | 34375.992            | 2751.716             | 33470.747            |

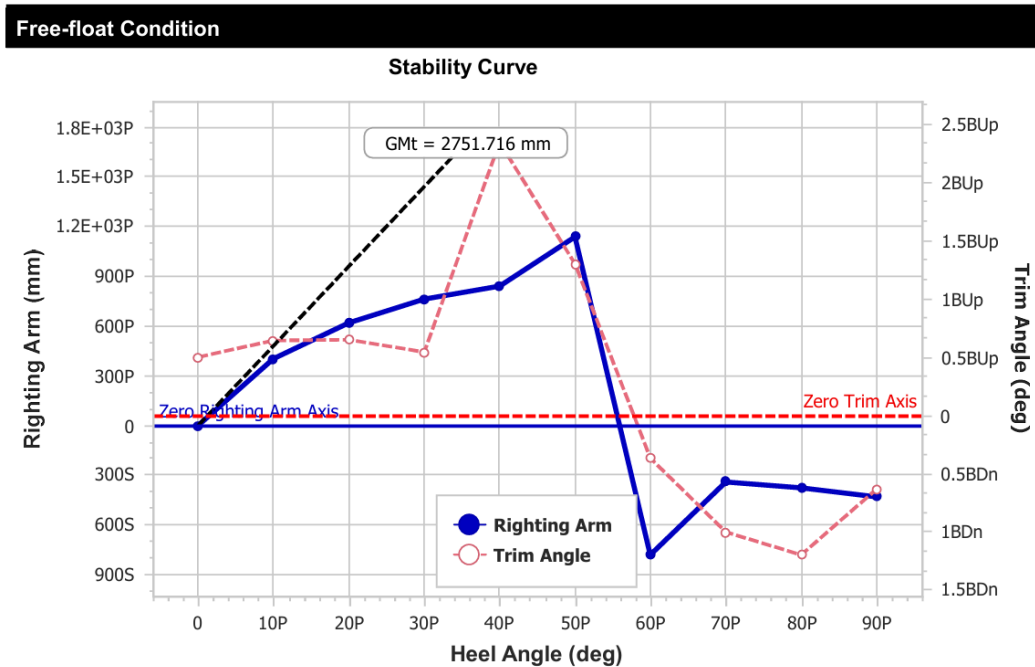
| Load Case            | C <sub>b</sub> | C <sub>p</sub> | C <sub>wp</sub> | C <sub>x</sub> | C <sub>ws</sub> | C <sub>vp</sub> |
|----------------------|----------------|----------------|-----------------|----------------|-----------------|-----------------|
| Free-float Condition | 0.408          | 0.000          | 0.795           | 0.000          | 3.565           | 0.513           |

#### Notes

**Figure 21 : Basic Hydrostatics Information**

A hydrostatic analysis was carried out using the Orca3D plug-in within Rhino 3D, and the key stability and flotation characteristics of the hull were obtained under the free-float condition. The model was evaluated with an applied weight of 15,000 kgf and a preset trim of 0.5°. Based on these inputs, Orca3D calculated a sinkage of 210.735 mm and confirmed a heel angle of 0°. The resulting displacement was 14,999.995 kgf, with the longitudinal centre of buoyancy located at 10,008.219 mm and the vertical centre of buoyancy at 94.790 mm. The waterplane area was found to be 44.444 m<sup>2</sup>, and the longitudinal centre of flotation was positioned at 9,455.272 mm. Hydrostatic stability indicators such as BM<sub>t</sub> (3656.960 mm), BM<sub>I</sub> (34375.992 mm), and GM<sub>t</sub> (2751.716 mm) were also obtained, providing an overview of the vessel's transverse and longitudinal stability. Additionally, form coefficients including block

coefficient ( $C_b = 0.408$ ), waterplane coefficient ( $C_{wp} = 0.795$ ), and prismatic coefficients were calculated to describe the hull's geometric efficiency.



**Figure 22 : Heel Angle vs righting arm and trim angle**

By examining the stability curve obtained under the free-float condition, the vessel demonstrates a stable behaviour throughout the range of applied heel angles. The righting arm increases steadily from  $0^\circ$  and reaches its maximum value at approximately  $50^\circ$  of heel, indicating the point at which the hull generates the greatest restoring moment. Beyond this angle, the righting arm begins to decrease; however, it remains positive up to  $90^\circ$ , showing that the vessel maintains its ability to return to an upright position without capsizing. The trim angle follows a similar trend, initially rising with heel before reducing after its peak around the mid-range of angles. The positive GMt value of 2751.716 mm further confirms that the vessel possesses adequate initial transverse stability. Overall, the plot illustrates that the hull is capable of withstanding significant heeling moments while preserving a positive righting response across the evaluated heel range.

| Heel (deg) | Trim (deg) | Righting Arm (mm) | Righting Moment (kgf-m) | Righting Energy (m-rad) | Notes |
|------------|------------|-------------------|-------------------------|-------------------------|-------|
| 0.00       | 0.50BUp    | 0.000             | 0.000                   | 0.000                   |       |
| 10.00P     | 0.64BUp    | 401.996P          | 6029.922P               | 0.035                   |       |
| 20.00P     | 0.65BUp    | 618.458P          | 9276.851P               | 0.124                   |       |
| 30.00P     | 0.54BUp    | 762.772P          | 11441.567P              | 0.245                   |       |
| 40.00P     | 2.34BUp    | 840.455P          | 11277.942P              | 0.385                   |       |
| 50.00P     | 1.30BUp    | 1138.334P         | 5826.769P               | 0.557                   |       |
| 60.00P     | 0.37BDn    | 781.269S          | 3172.718S               | 0.588                   |       |
| 70.00P     | 1.00BDn    | 341.172S          | 3634.776S               | 0.490                   |       |
| 80.00P     | 1.20BDn    | 376.555S          | 5169.093S               | 0.428                   |       |
| 90.00P     | 0.63BDn    | 430.412S          | 5329.538S               | 0.357                   |       |

**Figure 23 : Table of intact stability**

## **4 General Arrangement Plan Drawings**

### **4.1 General Specifications**

- Overall Length (LOA) : 15 m
- Beam (B) : Approx. 5 meters
- Draft (D) : Approx. 1.0 meters
- Maximum Speed : 30 knots
- Propulsion : Preferably 02 x Waterjet Propulsion System
- Material : FRP
- Range : 250 NM at cruising speed (30 knots)

## 4.2 Accommodation and Layout

### 4.2.1 Main Deck

- Cockpit / Helm Station: Forward control area with helm console, navigation displays, communication systems, and seating for the coxswain and crew. Designed for clear visibility and quick maneuvering during emergency missions.
- Medical Treatment Cabin: Central enclosed space with one secured stretcher bed, a foldable backup stretcher, and seating for up to six medical personnel or stable patients. Includes oxygen storage points, mounting brackets for portable equipment, and sufficient circulation space for treatment during transit.
- Patient Transfer Access: A wide side door or foldable ramp that enables smooth stretcher loading from jetties or small quays, ensuring safe and rapid embarkation.
- Medical Storage Area: Compact cabinets and overhead compartments for defibrillators, suction units, ventilators, and essential consumables, secured to prevent movement during high-speed runs.

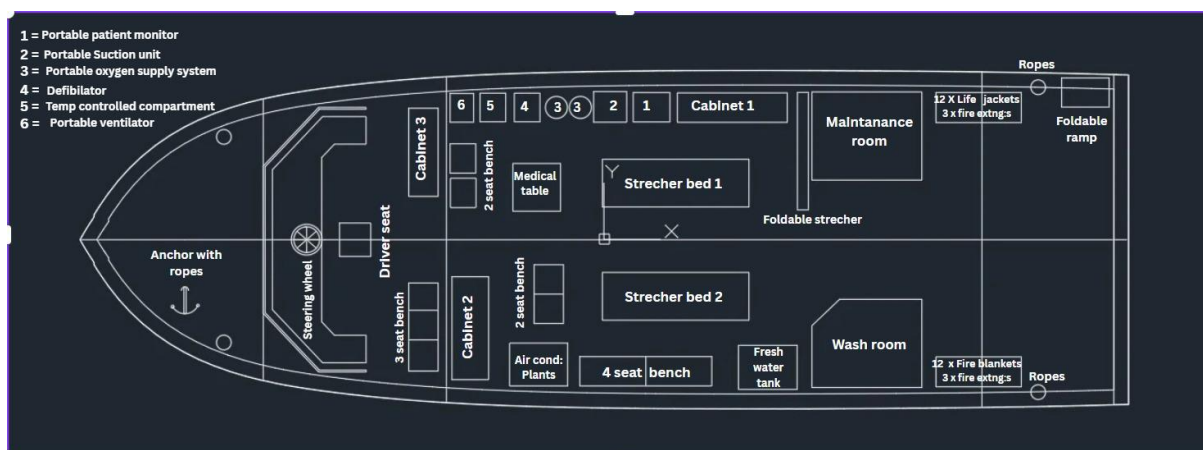
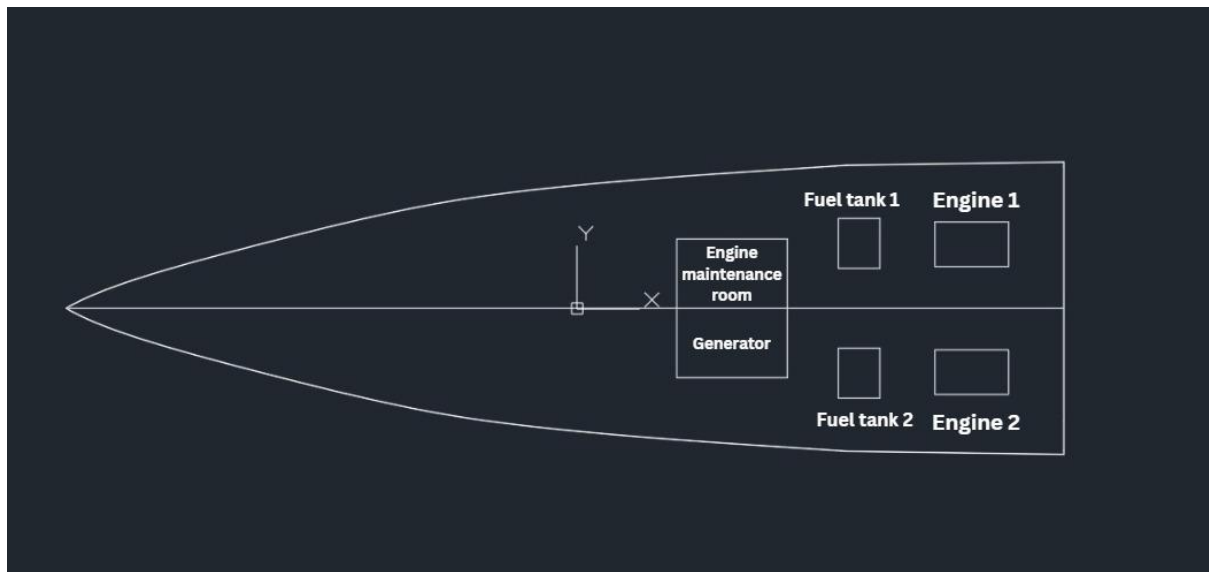


Figure 23 : Main Deck's GA Plan

#### 4.2.2 Lower Deck

- **Engine Room:** Dedicated compartment housing the main engines, waterjet drives, cooling systems, and ventilation ducts. Provides service access around machinery for routine inspections and emergency maintenance.
- **Fuel Tank Compartment:** Integrated FRP fuel reservoir designed to provide at least 4 hours of endurance at maximum speed. Includes internal baffles, inspection ports, and compliant venting for safety in rough or shallow-water operations.
- **Technical Systems Bay:** Area assigned for electrical systems, batteries, inverters, bilge pumps, and control manifolds, positioned to keep sensitive equipment isolated from engine heat and vibration.
- **Lower Deck Storage:** Secured stowage for additional oxygen cylinders, medical supply packs, and backup gear, helping maintain an uncluttered main medical cabin.



**Figure 24 : Lower Deck's GA Plan**

### **4.2.3 Functional and Medical Features**

#### **Exterior**

- **Rapid Boarding Access:** Side door or foldable loading ramp designed for smooth stretcher transfer from jetties and small quays.
- **Safe Working Deck:** Non-slip walkway surfaces with grab rails for secure movement during rough conditions.
- **Emergency Lighting:** High-visibility exterior LED lights for night operations, patient transfer, and rescue coordination.

#### **Interior**

- **Medical Treatment Space:** Dedicated cabin with a secure stretcher bed, foldable stretcher, and seating for medical staff to monitor patients during transit.
- **Equipment Mounting Points:** Brackets and shelves for portable oxygen units, ventilators, defibrillators, monitors, and suction devices.
- **Climate-Controlled Cabin:** Air-conditioned interior to maintain patient comfort in tropical environments.
- **Sanitized & Durable Finishes:** Use of medical-grade, easy-to-clean materials for hygiene and infection control.

### **4.2.4 Amenities & Systems**

- **Electrical Systems:** Dual-voltage onboard power with inverters and battery backup dedicated to critical medical equipment.
- **Water Systems:** Freshwater tank with basic washing facilities for medical use and sanitation. Emergency hand-wash and equipment-cleaning provisions included.
- **Navigation & Communication:** GPS, Radar, AIS, VHF radio, and loudhailer system for coordinated emergency operations.

- Medical Support Systems: Dedicated oxygen storage, secured medical cabinet space, and optional small refrigerator for medication requiring cooling.

#### 4.2.5 Safety & Compliance

- Safety Equipment: Fire extinguishers, fire blanket, life jackets for all occupants, automatic bilge pumps, and emergency beacons.
- Regulatory Compliance: Full adherence to Sri Lanka Small Craft Code, including stability, lifesaving equipment, and operational safety requirements for medical transport vessels.

## 5 Weight and Weight Center Estimation

The individual weights and the centres of gravity of each component and structural part were listed and used to calculate their moments. Based on these calculations, the fixed centre of gravity of the entire ship was finally determined.

|                           |     |
|---------------------------|-----|
| Length overall (m)        | 15  |
| Beam (m)                  | 5   |
| Deck height at bow (m)    | 1.9 |
| Deck height at transom(m) | 1.6 |

The port side was considered as the positive side and the starboard side was considered as the negative side when calculating moments.

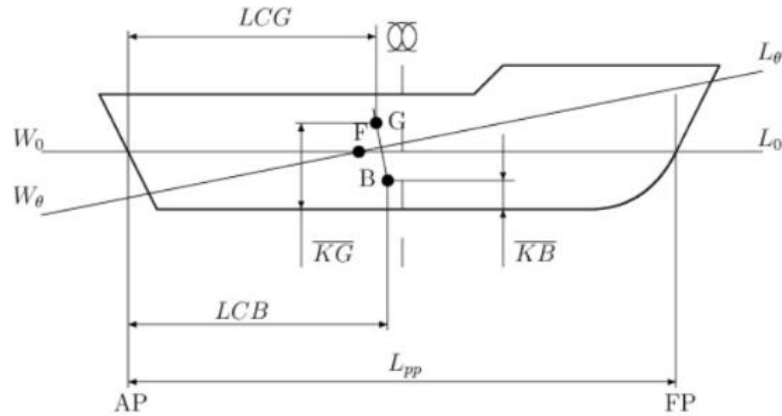
| Description                                                      | Full load weights(kg) | LCG from Aft perp (m) | LMOM (kgm)       | VCG from baseline (m) | VMOM (kgm)     | TCG from centerline (m) | TMOM (kgm)   |
|------------------------------------------------------------------|-----------------------|-----------------------|------------------|-----------------------|----------------|-------------------------|--------------|
| <b>Light craft weight</b>                                        |                       |                       |                  |                       |                |                         |              |
| Hull (Material - FRP)                                            | 5800                  | 10                    | 58000            | 1                     | 5800           | 0                       | 0            |
| All Longitudinal Frames                                          | 700                   | 7.5                   | 5250             | 0.85                  | 595            | 0                       | 0            |
| All Transverse frames                                            | 700                   | 7.5                   | 5250             | 0.85                  | 595            | 0                       | 0            |
| All Bulkheads                                                    | 600                   | 7.5                   | 4500             | 0.85                  | 510            | 0                       | 0            |
| Electrical Accessories                                           | 150                   | 6.17                  | 925.5            | 1.7                   | 255            | 0                       | 0            |
| Deck with frames and AL cheq plate                               | 1000                  | 6.5                   | 6500             | 0.85                  | 850            | 0                       | 0            |
| Deck pillars                                                     | 200                   | 7.5                   | 1500             | 2.5                   | 500            | 0                       | 0            |
| Fuel Tanks                                                       | 150                   | 5                     | 750              | 0.7                   | 105            | 0                       | 0            |
| Sewage Tank                                                      | 60                    | 6                     | 360              | 0.7                   | 42             | 0                       | 0            |
| Fresh Water Tank                                                 | 80                    | 4.92                  | 393.6            | 1.9                   | 152            | 1.73                    | 138.4        |
| Superstructure With fiber glass                                  | 800                   | 7.5                   | 6000             | 3                     | 2400           | 0                       | 0            |
| 4 Passenger Chairs (medical) with mounts                         | 30                    | 6.58                  | 197.4            | 1.7                   | 51             | 1.79                    | 53.7         |
| 2 Passenger Chairs (medical) with mounts                         | 15                    | 7.91                  | 118.65           | 1.7                   | 25.5           | 0.74                    | 11.1         |
| 2 Passenger Chairs (medical) with mounts                         | 15                    | 9.08                  | 136.2            | 1.7                   | 25.5           | -0.9                    | -13.5        |
| 3 Personal Chairs (crew) with mounts                             | 22.5                  | 9.62                  | 216.45           | 1.7                   | 38.25          | 1.16                    | 26.1         |
| 1 Coxswain Chair with mounts                                     | 7.5                   | 10.54                 | 79.05            | 1.7                   | 12.75          | 0                       | 0            |
| 2 x Main stretcher system (fully equiped secured emergency beds) | 180                   | 6.17                  | 1110.6           | 1.7                   | 306            | 0                       | 0            |
| Foldable stretcher                                               | 15                    | 4.55                  | 68.25            | 1.7                   | 25.5           | -1.8                    | -27          |
| Portable oxygen supply system                                    | 60                    | 7.47                  | 448.2            | 1.7                   | 102            | -1.8                    | -108         |
| Defibrillator                                                    | 5                     | 8.68                  | 43.4             | 1.7                   | 8.5            | -1.8                    | -9           |
| Portable suction unit                                            | 7                     | 7.06                  | 49.42            | 1.7                   | 11.9           | -1.8                    | -12.6        |
| Portable patient monitor                                         | 10                    | 6.5                   | 65               | 1.7                   | 17             | -1.8                    | -18          |
| Portable ventilator                                              | 10                    | 9.08                  | 90.8             | 1.7                   | 17             | -1.8                    | -18          |
| Shock proof cabinet space (Essential items)                      | 80                    | 5.4                   | 432              | 1.7                   | 136            | -1.8                    | -144         |
| Shock proof cabinet space (Excess storage)                       | 80                    | 8.98                  | 718.4            | 1.7                   | 136            | 1.2                     | 96           |
| Shock proof cabinet space (Control room)                         | 80                    | 9.62                  | 769.6            | 1.7                   | 136            | -1.19                   | -95.2        |
| Console With Equipments                                          | 120                   | 11.38                 | 1365.6           | 1.9                   | 228            | 0                       | 0            |
| 30 KVA Generator                                                 | 450                   | 9.3                   | 4185             | 0.7                   | 315            | 0                       | 0            |
| Air Conditioning Plants                                          | 120                   | 8.05                  | 966              | 1.7                   | 204            | 1.68                    | 201.6        |
| Roof Structure                                                   | 300                   | 7.5                   | 2250             | 2.8                   | 840            | 0                       | 0            |
| Bathroom items                                                   | 100                   | 3.55                  | 355              | 1.7                   | 170            | 1.41                    | 141          |
| Anchor with ropes                                                | 100                   | 12.34                 | 1234             | 1.7                   | 170            | 1.39                    | 139          |
| 3 Extinguisher Types with 1 Tank Each (Port side)                | 45                    | 2.04                  | 91.8             | 1.7                   | 76.5           | 1.8                     | 81           |
| 3 Extinguisher Types with 1 Tank Each (Starboard side)           | 45                    | 2.04                  | 91.8             | 1.7                   | 76.5           | -1.8                    | -81          |
| 12 x Fire blankets (Port side)                                   | 18                    | 2.04                  | 36.72            | 1.7                   | 30.6           | 1.8                     | 32.4         |
| 12 x Life Jackes (Starboard side)                                | 12                    | 2.04                  | 24.48            | 1.7                   | 20.4           | -1.8                    | -21.6        |
| Floodable ramp                                                   | 100                   | 0.72                  | 72               | 1.7                   | 170            | -2                      | -200         |
| Temperature controlled compartment for heat sensitive drugs      | 40                    | 7.8                   | 312              | 1.7                   | 68             | -1.8                    | -72          |
| Maintenance equipments (Basic toolkit etc.)                      | 60                    | 3.51                  | 210.6            | 1.7                   | 102            | -1.41                   | -84.6        |
| Medical table                                                    | 40                    | 8.05                  | 322              | 1.7                   | 68             | -0.71                   | -28.4        |
| <b>Light load condition weight</b>                               | <b>12407</b>          |                       | <b>105489.52</b> |                       | <b>15391.9</b> |                         | <b>-12.6</b> |
| <b>Dead weight</b>                                               |                       |                       | 0                |                       | 0              |                         | 0            |
| Swedge                                                           | 20                    | 6                     | 120              | 0.7                   | 14             | 0                       | 0            |
| Fuel weight                                                      | 900                   | 5                     | 4500             | 0.7                   | 630            | 0                       | 0            |
| Fresh Water weight                                               | 200                   | 4.92                  | 984              | 1.9                   | 380            | 1.73                    | 346          |
| Cargo                                                            | 100                   | 5.4                   | 540              | 1.7                   | 170            | -1.8                    | -180         |
| Passengers with crew (Max 12)                                    | 900                   | 6.17                  | 5553             | 1.7                   | 1530           | 0                       | 0            |
|                                                                  | 2120                  |                       | 11697            |                       | 2724           |                         | 166          |
| <b>Full load condition weight</b>                                | <b>14527</b>          |                       | <b>117186.52</b> |                       | <b>18115.9</b> |                         | <b>153.4</b> |

**Figure 25 : Weight centre Calculations**

The final centres of gravity of the craft are given below:

| Load condition          | Full load   | Light load   |
|-------------------------|-------------|--------------|
| LCG from transom (m)    | 8.066808013 | 8.502419602  |
| VCG from baseline (m)   | 1.24705032  | 1.24058193   |
| TCG from centerline (m) | 0.010559648 | -0.001015556 |

**Figure 26 : Locations of centre of gravities**



**Figure 27: Centres of gravity**

For good planing performance, the LCG of a planing hull is generally expected to be around 50% of the vessel's total length. In this design, the LCG from the transom is 8.07 m under full-load conditions and 8.50 m under light-load conditions. With an overall vessel length of 15 m, these values correspond to approximately 53.8% and 56.7% of the total length, respectively.

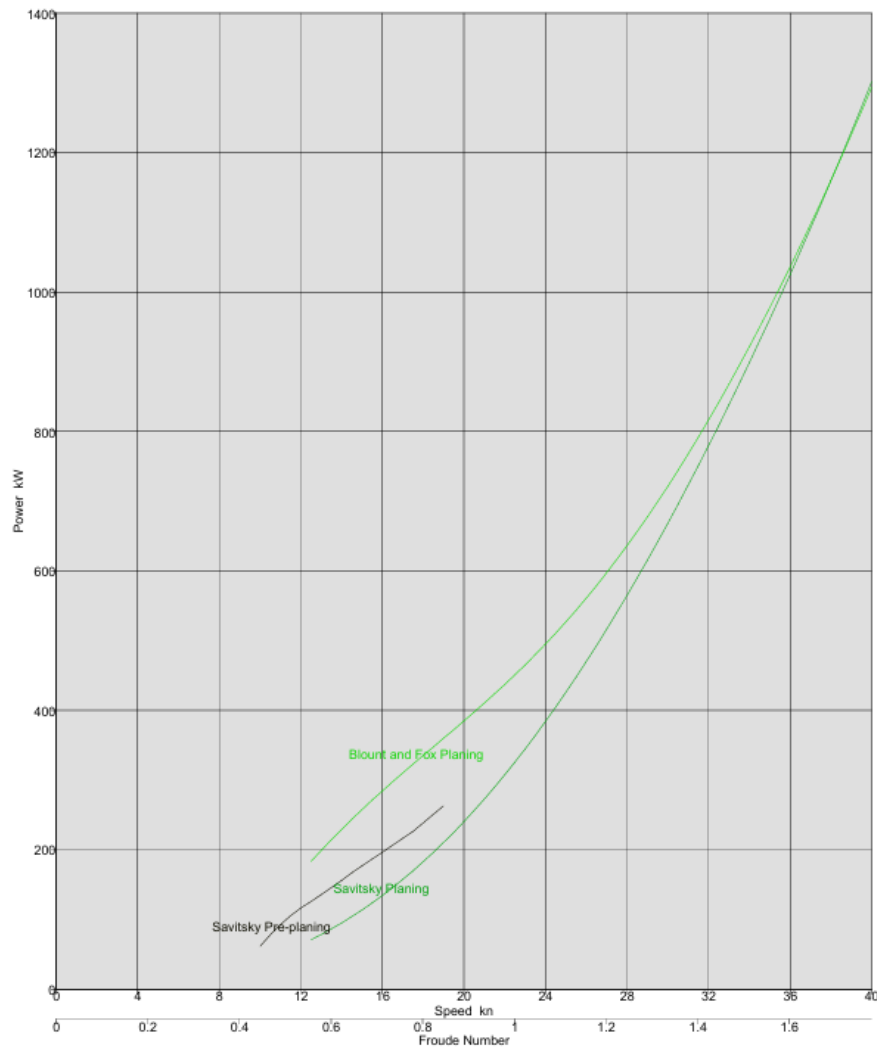
The TCG values are very close to zero (0.0106 m in full load and  $-0.0010$  m in light load), which indicates that the craft is expected to remain upright under fixed loading conditions. Further stability analysis of the ambulance craft was carried out using the Orca3D add-in.

## 6 Resistance and Power

|    | Speed<br>(kn) | Froude No.<br>LWL | Froude No.<br>Vol. | Savitsky<br>Pre-planing<br>Resist. (kN) | Savitsky<br>Pre-planing<br>Power (kW) | Savitsky<br>Planing<br>Resist. (kN) | Savitsky<br>Planing<br>Power (kW) | Blount and Fox<br>Planing<br>Resist. (kN) |    | Blount and Fox<br>Planing<br>Power (kW) |
|----|---------------|-------------------|--------------------|-----------------------------------------|---------------------------------------|-------------------------------------|-----------------------------------|-------------------------------------------|----|-----------------------------------------|
| 1  | 0.000         | 0.000             | 0.000              | --                                      | --                                    | --                                  | --                                | --                                        | 1  | --                                      |
| 2  | 1.000         | 0.044             | 0.103              | --                                      | --                                    | --                                  | --                                | --                                        | 2  | --                                      |
| 3  | 2.000         | 0.089             | 0.205              | --                                      | --                                    | --                                  | --                                | --                                        | 3  | --                                      |
| 4  | 3.000         | 0.133             | 0.308              | --                                      | --                                    | --                                  | --                                | --                                        | 4  | --                                      |
| 5  | 4.000         | 0.178             | 0.410              | --                                      | --                                    | --                                  | --                                | --                                        | 5  | --                                      |
| 6  | 5.000         | 0.222             | 0.513              | --                                      | --                                    | --                                  | --                                | --                                        | 6  | --                                      |
| 7  | 6.000         | 0.267             | 0.616              | --                                      | --                                    | --                                  | --                                | --                                        | 7  | --                                      |
| 8  | 7.000         | 0.311             | 0.718              | --                                      | --                                    | --                                  | --                                | --                                        | 8  | --                                      |
| 9  | 8.000         | 0.356             | 0.821              | --                                      | --                                    | --                                  | --                                | --                                        | 9  | --                                      |
| 10 | 9.000         | 0.400             | 0.923              | --                                      | --                                    | --                                  | --                                | --                                        | 10 | --                                      |
| 11 | 10.000        | 0.445             | 1.026              | 9.6                                     | 61.783                                | --                                  | --                                | --                                        | 11 | --                                      |
| 12 | 11.000        | 0.489             | 1.128              | 13.0                                    | 92.211                                | --                                  | --                                | --                                        | 12 | --                                      |
| 13 | 12.000        | 0.534             | 1.231              | 15.1                                    | 116.370                               | --                                  | --                                | --                                        | 13 | --                                      |
| 14 | 13.000        | 0.578             | 1.334              | 16.2                                    | 135.652                               | 9.4                                 | 78.459                            | 23.8                                      | 14 | 199.003                                 |
| 15 | 14.000        | 0.623             | 1.436              | 17.3                                    | 156.090                               | 10.6                                | 94.986                            | 25.4                                      | 15 | 229.057                                 |
| 16 | 15.000        | 0.667             | 1.539              | 18.3                                    | 176.942                               | 11.8                                | 113.597                           | 26.7                                      | 16 | 257.448                                 |
| 17 | 16.000        | 0.712             | 1.641              | 19.1                                    | 196.496                               | 13.1                                | 134.369                           | 27.6                                      | 17 | 284.354                                 |
| 18 | 17.000        | 0.756             | 1.744              | 19.8                                    | 216.248                               | 14.4                                | 157.370                           | 28.4                                      | 18 | 310.100                                 |
| 19 | 18.000        | 0.801             | 1.847              | 20.6                                    | 238.359                               | 15.8                                | 182.657                           | 29.0                                      | 19 | 335.103                                 |
| 20 | 19.000        | 0.845             | 1.949              | 21.5                                    | 262.783                               | 17.2                                | 210.273                           | 29.5                                      | 20 | 359.826                                 |
| 21 | 20.000        | 0.890             | 2.052              | --                                      | --                                    | 18.7                                | 240.250                           | 29.9                                      | 21 | 384.746                                 |
| 22 | 21.000        | 0.934             | 2.154              | --                                      | --                                    | 20.2                                | 272.607                           | 30.4                                      | 22 | 410.318                                 |
| 23 | 22.000        | 0.979             | 2.257              | --                                      | --                                    | 21.7                                | 307.350                           | 30.9                                      | 23 | 436.963                                 |
| 24 | 23.000        | 1.023             | 2.359              | --                                      | --                                    | 23.3                                | 344.472                           | 31.4                                      | 24 | 465.052                                 |
| 25 | 24.000        | 1.068             | 2.462              | --                                      | --                                    | 24.9                                | 383.955                           | 32.1                                      | 25 | 494.899                                 |
| 26 | 25.000        | 1.112             | 2.565              | --                                      | --                                    | 26.5                                | 425.769                           | 32.8                                      | 26 | 526.756                                 |
| 27 | 26.000        | 1.157             | 2.667              | --                                      | --                                    | 28.1                                | 469.872                           | 33.5                                      | 27 | 560.821                                 |
| 28 | 27.000        | 1.201             | 2.770              | --                                      | --                                    | 29.7                                | 516.217                           | 34.4                                      | 28 | 597.234                                 |
| 29 | 28.000        | 1.246             | 2.872              | --                                      | --                                    | 31.4                                | 564.746                           | 35.3                                      | 29 | 636.087                                 |
| 30 | 29.000        | 1.290             | 2.975              | --                                      | --                                    | 33.0                                | 615.333                           | 36.3                                      | 30 | 677.433                                 |
| 31 | 30.000        | 1.335             | 3.078              | --                                      | --                                    | 34.6                                | 668.115                           | 37.4                                      | 31 | 721.288                                 |
| 32 | 31.000        | 1.379             | 3.180              | --                                      | --                                    | 36.3                                | 722.836                           | 38.5                                      | 32 | 767.644                                 |
| 33 | 32.000        | 1.424             | 3.283              | --                                      | --                                    | 37.9                                | 779.508                           | 39.7                                      | 33 | 816.475                                 |
| 34 | 33.000        | 1.468             | 3.385              | --                                      | --                                    | 39.5                                | 838.090                           | 40.9                                      | 34 | 867.750                                 |
| 35 | 34.000        | 1.513             | 3.488              | --                                      | --                                    | 41.1                                | 898.553                           | 42.1                                      | 35 | 921.433                                 |
| 36 | 35.000        | 1.557             | 3.590              | --                                      | --                                    | 42.7                                | 960.885                           | 43.4                                      | 36 | 977.495                                 |
| 37 | 36.000        | 1.602             | 3.693              | --                                      | --                                    | 44.3                                | 1025.096                          | 44.7                                      | 37 | 1035.918                                |
| 38 | 37.000        | 1.646             | 3.796              | --                                      | --                                    | 45.9                                | 1091.213                          | 46.1                                      | 38 | 1096.699                                |
| 39 | 38.000        | 1.691             | 3.898              | --                                      | --                                    | 47.4                                | 1159.286                          | 47.5                                      | 39 | 1159.850                                |
| 40 | 39.000        | 1.735             | 4.001              | --                                      | --                                    | 49.0                                | 1229.383                          | 48.9                                      | 40 | 1225.402                                |
| 41 | 40.000        | 1.780             | 4.103              | --                                      | --                                    | 50.6                                | 1301.590                          | 50.3                                      | 41 | 1293.399                                |

**Figure 28 : Speed vs Resistance power and Power table**

According to the project guidelines, a design speed of 30 knots was selected for the ambulance craft. Resistance and power requirements were evaluated over a speed range of 0 - 35 knots using MAXSURF Resistance. Froude numbers were used to identify the hull's transition from displacement to semi-planing and full planing regimes.



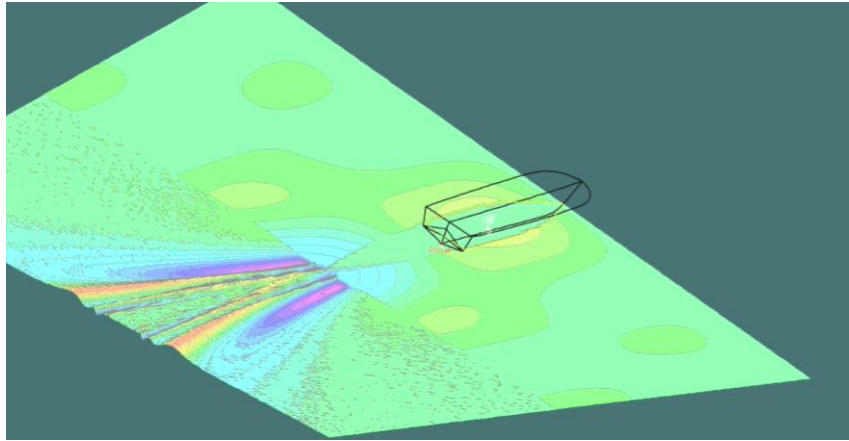
**Figure 29 : Power vs Speed graph**

The Savitsky method was used to predict resistance and power in both pre-planing and full planing conditions. Additional validation was carried out using the Blount & Fox planing resistance model. The results from these methods were then used to estimate the required engine power and to confirm the hull's ability to achieve the target operating speed.

The software allows the inclusion of overall efficiency. Therefore, an initial total efficiency of 70% was considered.

From the Savitsky method, the planing power requirement was 615.399 kW, with a corresponding resistance force of 33 kN.

From the Blount & Fox method, the planing power requirement was 677.433 kW, with a resistance force of 30 kN.



**Figure 30 : Wave pattern visualization (using MAXSURF Resistance CONNECT Edition x64 software)**

Between these two predictions, the higher value of 677 kW was taken as the maximum calm water power requirement. To account for rough sea operation, a typical service factor  $C \approx 1.1$  (approximately a 10% sea margin) was applied. Accordingly, the required power in rough conditions is:

$$P_{\text{rough}} = C \times P_{\text{calm}} = 1.1 \times 677.433 = 745.18 \text{ kW}$$

Considering losses in the shafting, gearbox, and propeller, along with the rough service margin, the overall system efficiency was taken as 60%. This value is consistent with typical efficiencies for high-speed craft and small ships, which generally fall in the 50 - 60% range, and is therefore representative of real-world operating conditions.

To accommodate the required 745 kW of power, two 600 hp inboard Cummins marine engines were selected. Each engine provides approximately 447 kW, giving a total installed power of 894 kW, which is sufficient and provides an adequate margin over the required power.



**Figure 31 : 600 hp inboard Cummins marine engine**

**Key attributes**

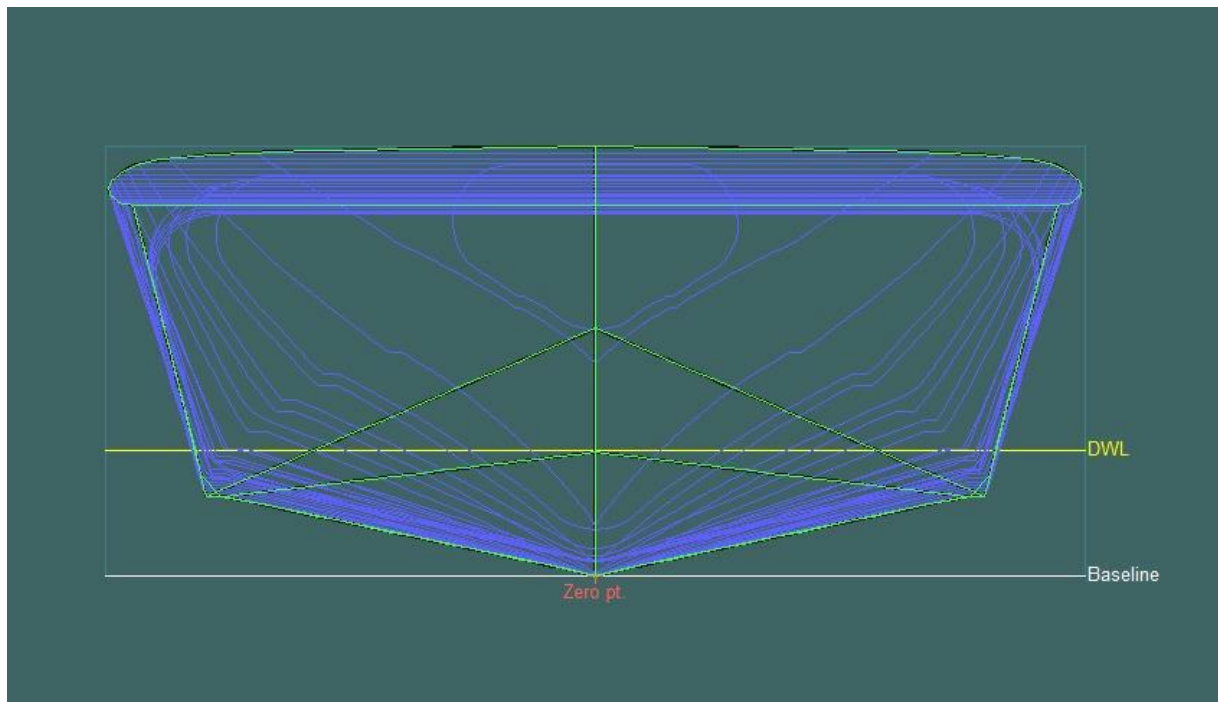
|                             |                                 |                 |                 |
|-----------------------------|---------------------------------|-----------------|-----------------|
| Condition                   | New                             | Condition       | New             |
| Engine                      | 4 Stroke                        | Fuel Type       | Diesel          |
| Cylinders                   | 6                               | Engine Position | Inboard         |
| Dimensions (L x W x H) (mm) | 2.1*1.0*2.0m                    | Engine Model    | Cumins KTA19-M3 |
| Engine Power                | 600HP/447KW                     | Engine Speed    | 1800RPM         |
| Warranty                    | 1 Year                          | Starting system | Electric        |
| Fuel Consumption            | 0.25L/kw.h                      | Condition       | New             |
| Standard Parts              | Heat Exchanger & Sea Water Pump | Option          | Advance GearBox |
| Keyword                     | Marine Engine 600HP             |                 |                 |

**Figure 32 : Key attributes of the engine**

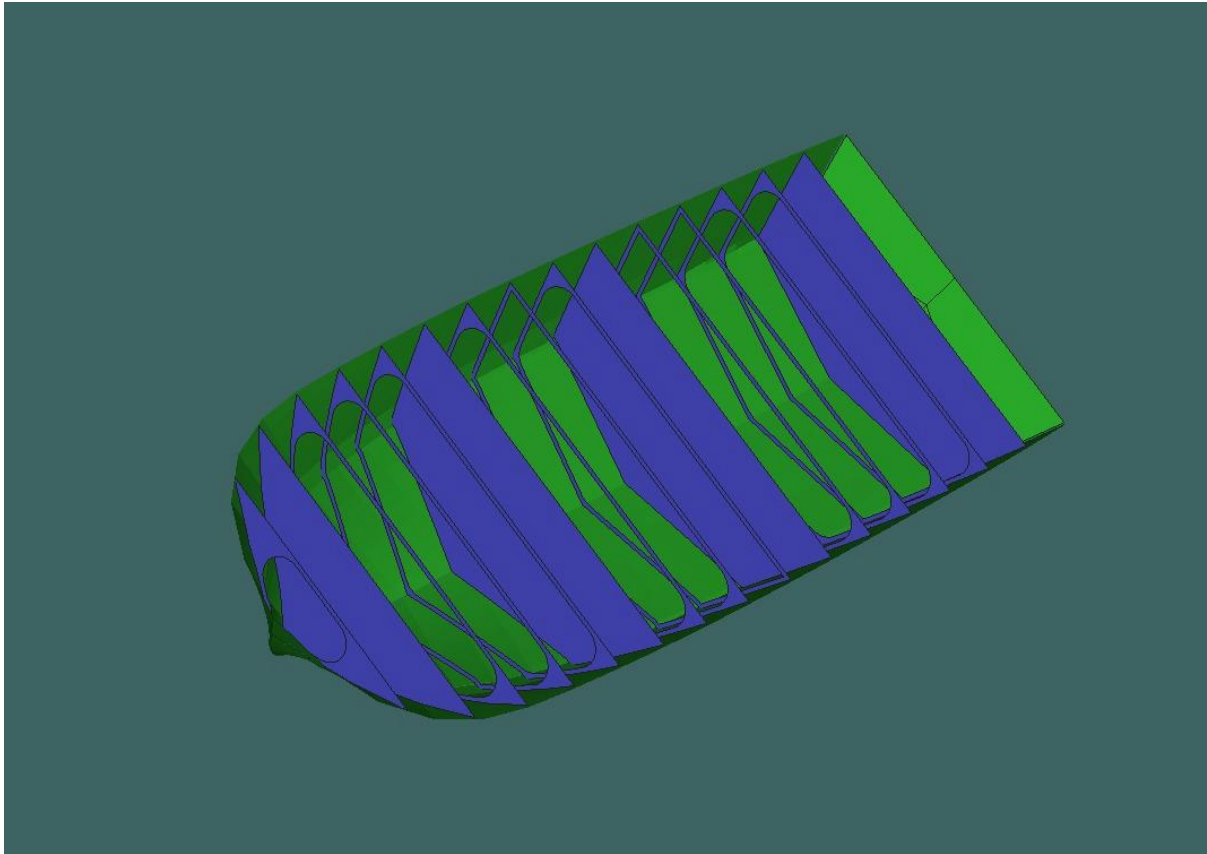
## 7 Subdivisions and Freeboard Comparison

### 7.1 Bulkheads

Bulkheads were incorporated to subdivide the hull and provide structural support and basic watertight integrity. Since the selected parent hull arrangement included five bulkheads, the same number was adopted for the proposed application without performing additional detailed calculations at this stage. This approach ensures consistency with the reference design while providing a practical and proven internal subdivision for the vessel.



**Figure 33 : Front View of Bulkheads**



**Figure 34 : Bulkheads Designs**

## **8 Cost Analysis**

Cost estimation was carried out to obtain an approximate budget for the proposed craft during the early design stage. Major cost components such as hull construction, propulsion system, superstructure, and onboard equipment were considered to ensure a realistic overall estimate. The results provide a basis for comparing design alternatives and assessing the economic feasibility of the proposed vessel.

**Table 1 : Cost Estimation**

| Material and Component                        | Quantity  | Cost Calculation(USD) |
|-----------------------------------------------|-----------|-----------------------|
| Total Material cost                           |           | 460000                |
| Main stretcher system (secured emergency bed) | 2         | 24,000                |
| Foldable stretcher                            | 1         | 18                    |
| Spinal board                                  | 1         | 149                   |
| Foldable Ramp                                 | 1         | 80                    |
| Patient transfer sheet                        | 1         | 30                    |
| Wheelchair (foldable)                         | 1         | 130                   |
| Portable Oxygen supply system                 | 1         | 3200                  |
| Portable Ventilator                           | 1         | 1200                  |
| Portable suction unit                         | 1         | 65                    |
| Portable patient monitor                      | 2         | 450                   |
| Defibrillator                                 | 1         | 650                   |
| Manual resuscitator (Ambu bag)                | 1         | 6                     |
| Nebulizer kit                                 | 1         | 13                    |
| Storage cabinets                              | 1         | 18                    |
| Laryngoscope set                              | 1         | 35                    |
| First aid kit                                 | 1         | 12                    |
| Basic emergency medicines box                 | 1         | 25                    |
| portable extinguishers*3                      | 3         | 225                   |
| fire blanket                                  | 1         | 25                    |
| Life jackets                                  | 12        | 1080                  |
| Lifebuoys                                     | 12        | 432                   |
| Engine                                        | 2         | 52000                 |
| Freshwater tank                               | 200litres | 250                   |
| Generator                                     | 40Kw      | 17000                 |

|                                         |  |         |
|-----------------------------------------|--|---------|
| <b>Total Material &amp; equipment</b>   |  | 561093  |
| Labour and engineering cost             |  | 36000   |
| Design and Engineering Fees             |  | 135000  |
| <b>Total initial construction costs</b> |  | 171000  |
| Total fuel cost(1000hr)                 |  | 264600  |
| Engine and Hull maintenance             |  | 36600   |
| System upkeep                           |  | 5000    |
| Interior upkeep                         |  | 8000    |
| Total Docking and Storage               |  | 6200    |
| Total insurance                         |  | 10500   |
| Total Crew and Staff Costs              |  | 60000   |
| <b>Total operational cost per year</b>  |  | 390900  |
| Total Cost Estimation                   |  | 1122993 |